Lab 4: Functional Dependencies

Research Lab Management System (RLMS)

Data Management · Fall 2026 · UM6P – College of Computing
Professor: Karima Echihabi · Program: Computer Engineering
Group: [Group name / number]

---

1. Team Members and Contributions

Member	Responsibility
Ahmad KHRISS	FDs of People; coordination and final report
Souhayl MOUJAHID	FDs of Laboratories, Projects, Budgets
Adam NAJI	FDs of Equipment, Certifications, Reservations, Maintenance, Calibration
Ilyas Youssef MOUHRI	FDs of Consumables, Suppliers, Stock; conditional FDs and assumptions

2. Functional Dependencies of the RLMS Schema

The FDs are derived from the relational schema of Lab 2 (itself mapped from the ER diagram of Lab 1). In every relation, the primary key determines all other attributes. Notation: $X \to Y$ means that the value of X determines the value of Y.

2.1 People

Relation	Functional dependency	Justification
Person	pid → firstName, lastName, email, affiliation, startDate, endDate	pid is the primary key. A person has one name, one email, one affiliation and one start/end date.
Student	pid → program	pid is the primary key (and a foreign key to Person). A student is enrolled in one program (BSc, BEng, MSc or PhD).
Employee	pid → phone, office, supervisorPid	pid is the primary key. ReportsTo was absorbed into Employee: only the current supervisor is kept (PE5), so each employee has at most one supervisorPid (NULL for the Dean).
Faculty, Administrative, Technical	pid → position	pid is the primary key of each subclass. A staff member has one position.

2.2 Laboratories, Projects and Budgets

Relation	Functional dependency	Justification
Laboratory	labId → name, building, roomNumber, discipline, supervisorPid	labId is the primary key. Supervises was absorbed: every laboratory has exactly one Faculty supervisor.
ResearchProject	code → title, startDate, endDate, status	code is the primary key. A project has one title, one period and one status.
Participates	(pid, projectCode) → role	role is an attribute of the pair: a person has exactly one role in a given project (PR3–PR4). pid alone does not determine role, since a person may have different roles in different projects.
Budget	budgetLine → amountGranted, amountDisbursed, startDate, endDate, managerPid	budgetLine is the primary key. Manages was absorbed: every budget is managed by exactly one Academic.

2.3 Equipment, Certifications and Reservations

Relation	Functional dependency	Justification
EquipmentModel	modelId → commercialName, manufacturer, category, requiredEnvironment, trainingMandatory	modelId is the primary key. These are properties of the model, shared by all its units.
EquipmentUnit	serialNo → acquisitionDate, purchaseCost, status, portable, modelId, labId	serialNo is the primary key. InstanceOf and LocatedIn were absorbed: a unit has exactly one model and is currently in exactly one laboratory.
Certification	code → title, issuingAuthority, validityPeriod, safetyLevel	code is the primary key.
Holds	(pid, certCode) → issueDate, expirationDate, grade	One instance per (person, certification): only the latest award is kept (CE3–CE4). Dates and grade describe that award.
Reservation	resId → submissionTS, plannedStart, plannedEnd, purpose, status, madeByPid, projectCode, approverPid	resId is the primary key. MadeBy and For were absorbed (exactly one person and one project). Approves was absorbed with a nullable approverPid (at most one approver).

2.4 Maintenance and Calibration (weak entities)

Relation	Functional dependency	Justification
Maintenance	(serialNo, startTS) → endTS, type, description, cost, outcome, technicalPid	The key is the owner's key plus the partial key startTS. DoneBy was absorbed: each maintenance is done by exactly one Technical staff member.

Relation	Functional dependency	Justification
CalibrationRecord	(serialNo, calibDate) → calibrationType, result, nextDueDate, remarks	The key is the owner's key plus the partial key calibDate.

2.5 Consumables, Suppliers and Stock

Relation	Functional dependency	Justification
Consumable	consId → name, unitOfMeasure, hazardLevel, reorderThreshold	consId is the primary key.
Supplier	suppId → name, contactEmail, phone	suppId is the primary key.
Supplies	(suppId, consId, labId) → unitPrice	Ternary relationship: unitPrice depends on the exact Supplier–Consumable–Laboratory triple (CO4). No proper subset of the triple determines it.
Stocks	(labId, consId) → quantityOnHand, lastRestockDate, storageCondition, technicalPid, monitoringSince	A stock occurrence is identified by (labId, consId). Monitors (on the aggregation of Stocks) has exactly one Technical monitor per stock, so technicalPid and monitoringSince are stored here (CO8).
Consumes	(resId, labId, consId) → quantityUsed	quantityUsed depends on the reservation and on the specific laboratory stock the quantity is taken from (CO9–CO10).

2.6 Relations with only trivial FDs

The following relations have all their attributes in the primary key, so they have no non-trivial FD: Academic, NonAcademic, Advises, Attached, Requires, FundsPrj, FundsLab, Reserves.

2.7 Conditional FDs

The Lab 2 schema declares no `UNIQUE` clause, so the FDs below are **not guaranteed by the schema**. They hold only if the stated assumption is true.

Relation	Possible FD	Assumption required
Person	email → pid (hence email → all Person attributes)	Email is unique per person.
Supplier	contactEmail → suppId	The contact email is unique per supplier.
EquipmentModel	(commercialName, manufacturer) → modelId	A model is identified by its commercial name and manufacturer.
Laboratory	(building, roomNumber) → labId	A room hosts only one laboratory.

Relation	Possible FD	Assumption required
Holds	(certCode, issueDate) → expirationDate	expirationDate = issueDate + validityPeriod (from Certification).

3. Key Challenges and Resolutions

- Rules that are not FDs (no self-reporting, reservation before use, certified users only) are left to triggers or CHECKs, as in Lab 2.
- `unitPrice` depends on the full (supplier, consumable, laboratory) triple, and a supply may exist without stock, so Supplies and Stocks stay separate.
- The relationnal schema declares no `UNIQUE` clause, so FDs such as email → pid are listed as conditional.

4. Justification of Conversion Choices

- ISA: one relation per class; each subclass key is also a foreign key to its parent.
- Weak entities: owner key + partial key, with `ON DELETE CASCADE`.
- Monitors is stored in Stocks (exactly one monitor per stock); Consumes references `Stocks(labId, consId)` to keep the supplying laboratory.

5. Assumptions

- Only the current supervisor, location and latest certification award are kept.
- The person with the highest authority has a NULL `supervisorPid`; a NULL `approverPid` means no approval is needed.